# TOUBAL Mohamed Sahnoune

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GitHub LinkedIn Portfolio

## About Me

Third-year AI Engineering student specializing in Machine Learning, Computer Vision, and Natural Language Processing. Experienced in building end-to-end AI systems, from training advanced **PyTorch** models to deploying scalable full-stack applications.

## Education

**ENSIA — National Higher School of Artificial Intelligence** *Expected 2028*
Third-Year Student
**Relevant Coursework**: Machine Learning, Data Mining, Mobile Dev, Web Dev, Data Structures & Algorithms I/II, AI, OOP, DB Systems, Linux, Time Series Analysis, Operational Research, Optimization, Networks, Security, OS.

## Experience

**Office Automation & IT Intern** — Ministry of Justice, Directorate General of Human Resources *Sep 2026*
- Drafted official letters, notices and reports in **Word**; registered incoming mail and routed it to the relevant units.
- Built **Excel** tracking tables with formulas; digitised and archived records following confidentiality and IT-security rules.

## Certifications

**Associate Data Scientist in Python** — DataCamp *2026*
Data manipulation, exploratory analysis, and supervised machine learning in Python. Credential

## Technical Skills

**AI & Machine Learning**
- **Computer Vision**: Image Classification, Face Recognition, Convolutional Neural Networks (CNNs).
- **Natural Language Processing**: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) Pipelines.
- **Machine Learning**: Federated Learning, Feature Engineering, Data Preprocessing, End-to-End ML Pipelines.
- **Frameworks & Libraries**: **PyTorch**, Scikit-learn, Pandas, NumPy.

**Software Engineering & Development**
- **Architecture**: Clean Architecture, BLoC/Cubit, Repository Pattern, Modular Application Design.
- **Programming**: Python, C++, Dart, JavaScript/TypeScript, SQL, Bash.
- **Full-Stack Development**: **Flutter**, React, Next.js, Express, PostgreSQL, **PostGIS**.
- **Systems & Tools**: On-Device AI Inference, Git, Linux, API Development, Scalable Deployment.

## Projects

**StreetWatch: Geospatial AI-Powered Road Damage Reporting** *2026*
- Developed a patch-based **CNN** for road damage classification, achieving **94% accuracy** with a 2-second inference time on mobile devices.
- Implemented clean architecture using the Repository pattern and BLoC in **Flutter** for robust **on-device AI** integration.
- Built a web dashboard with **PostGIS** that pairs **GPS coordinates** with AI-generated reports for spatial analysis and real-time mapping of infrastructure issues.

**Fantasy Premier League Prediction Pipeline** — fpl-assistant-ten.vercel.app *2025 – 2026*
- Built an end-to-end ML pipeline over **10 seasons** (254k player-matches), predicting weekly points to **1.0 MAE** on **48k** held-out rows from two unseen seasons.
- Trained per-position **LightGBM** models on **74** engineered features, beating a rolling-average baseline by **+0.12–0.15 $R^2$** under season-level holdout validation.
- Picked squads via **integer programming** (PuLP); the backtested XI beat the baseline by **+481 points** over a season.
- Deployed **FPL Assistant**, a web app driven by the pipeline's predictions, with **8 decision tools**: My Team, Transfer Studio, Captain & Form, Chip Advisor, Fixture Matrix, Comparison, Watchlist and News Wire.

**Drone Delivery Routing: Exact & Metaheuristic Optimisation** *2026*
- Solved a capacitated drone-routing problem with **load-dependent energy** exactly by **branch & bound**, cutting the search **1.5M → 429k nodes** and proving optimality to $n = 10$.
- Benchmarked GA, SA and **ALNS** on **242 literature instances** (CVRPLIB, Solomon): reproduced **all 74** published optima and showed ALNS > GA ($p = 2 \times 10^{-8}$) at a **3.3%** median gap on a 5 s budget.
- Shipped as a Python package (CLI, CI, **627 tests**) with QGroundControl mission export and a **10–20**× faster decoder.

## Languages

Arabic (Native) • English (Professional) • French (Professional, TCF C2)